[Edit: 1/9 - Course will be online until the end of January. Office hours will be on Zoom at https://washington.zoom.us/my/pisan]

Instructor: Dr. Yusuf Pisan, pisan@uw.edu UW1-260Q (425) 352-3741

Class: M/W 8:45-10:45. Week-1 on Zoom, the rest in UW1-051

Office Hours: Monday 1-2pm and Thursday 1-2pm. Signup via Canvas Calendar

Class Discord: invite link

Course Description

This sequenced course integrates mathematical principles with detailed instruction in computer programming. Topics include development of algorithms; algorithm analysis; object■oriented programming; abstract data types including trees, priority queues, graphs, and tables; regular expressions and context■free grammars. Prerequisites: CSS 342 with a grade of 2.0 or better.

Learning Objectives

- An understanding of trees, balanced trees, heaps, hash tables and their uses
- An understanding of the graph data structure and associated algorithms
- Ability to design and implement a complex object■oriented problem (using inheritance)
- An understanding of the formal notation for a programming language

Course Goals

Refining and extending the concepts and skills introduced in CSS 342, students develop competencies associated with problem■solving, design, testing, and programming techniques. Topics include ADTs, data structures, related algorithms, and object■oriented design & programming. Formal automata theory as it applies to programming languages is introduced. Good software engineering and algorithm analysis techniques are used throughout.

As with most technical courses, besides ability and motivation, it takes time to learn and master the subject. No one succeeds without practice! Expect to spend an average of 15 hours a week outside of class time for this course; some of you may spend more time, some less time.

Textbooks

- [Carrano] Frank M. Carrano and Timothy M Henry, Data Abstraction and Problem Solving with C++, Seventh edition, Addison-Wesley, 2017.

- [Cusack] Charles A. Cusack and David A. Santos, Active Introduction to Discrete Mathematics and Algorithms, Version 2.6.3, March 30, 2018.

- [Chen] William Chen, Discrete Mathematics

- [Rosen] Kenneth H. Rosen, Discrete Mathematics and Its Applications, Seventh Edition, , McGraw Hill, 2011.

Grading

- Exercises: 10%

- Projects: 40%

- Midterm: 25%

- Final: 25%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% of the possible points will receive a 2.0 grade in this course. Students with 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

Assignment Submission: Exercises will be graded as Complete/Incomplete. You can miss up to 3 exercises without penalty.

For projects, you can can submit one and only one project 24 hours late without any penalty. If you are using your 24 hour extension, post a brief explanation to "Using Extension" assignment.

Other than when working in pairs, talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct: http://www.washington.edu/cssc/for-students/student-code-of-conduct/ I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: We will use discord as an extension of the classroom. Use a meaningful nickname and act professionally. If your question can be answered publicly by me or by a classmate, post it to discord. Use the office hours for complex issues or topics you are struggling with.

Use your UW email rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, come and talk to me. If I don’t know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Course Material: Lecture notes and other material will be posted to Canvas under "Files". See List of Topics for detailed list of topics and additional references.

See the School of STEM Course Policies, which covers:

- Academic Integrity
- Access and Accommodations
- Classroom Emergency Preparedness
- For Our Veterans
- Grade of Incomplete
- Inclement Weather
- Parenting Resources

- Religious Accommodations

- Respect for Diversity

- Student Support Services

- Surviving Sexual and Relationship Violence

- Wonder How to Address Faculty?

Course Calendar

Week	Tuesday/Thursday	Notes

	3 Jan 5 Jan	Introduction, C++ Review	See List of Topics for readings and additional references Multiple exercises due (email, discord, unix, github, ...)
	10 Jan 12 Jan	Trees	**SVG Project**
	17 Jan 19 Jan	Graphs	17 Jan, Martin Luther King Day **BST Project**
	24 Jan 26 Jan	N-ary Trees	
	31 Jan 2 Feb	Design Patterns	**Graph Project**
	7 Feb 9 Feb	Review **Midterm**	
	14 Feb 16 Feb	Hash Table	15 Feb, Presidents' Day **LeetCode Project**

	21 Feb 23 Feb	Polymorphism	
	28 Feb 2 Mar	Theory of Computation	**Movies Design Project**
	7 Mar 9 Mar	Turing Machine	**Movies Project**
	14 Mar 16 Mar	**Final - 14 Mar**	No classes during finals week